Gilberto G. **Briscoe-Martinez**

PhD Candidate, Computer Science · NASA Space Technology Graduate Researcher
Boulder, CO, USA
gilbert.martinez675@gmail.com | gilberto.martinez.phd | ggb367 | gbmartinez | Google Scholar

Research Summary

I am a roboticist building truly independent robots. My PhD research gives robots a sense of self-understanding, so they autonomously adapt their behavior to keep working when they, or the environment around them, change: systems that don't shut down when things go wrong, but find safe, creative ways to finish the task without human rescue. I apply this where failure is costly and intervention is hard: long-term reliability for NASA space missions, safe contact-aware manipulation with the ROCC Lab, and intelligent package sorting at Plus One Robotics, where robots visually reason about failed actions, formulate recovery plans, and coordinate whole fleets of arms and conveyors to keep moving.

Education

University of Colorado Boulder — *Boulder, CO, USA*

PhD in Computer Science — *May 2027 (expected)*

- Advisor: Prof. Alessandro Roncone, HIRO Group.
- Research: fail-active manipulation, embodiment-aware policies, and contact-rich control.
- Funded by the NASA Space Technology Graduate Research Opportunity (NSTGRO) fellowship.

The University of Texas at Austin — *Austin, TX, USA*

B.S. in Aerospace Engineering — *May 2021*

- Undergraduate research in the **Human Centered Robotics Lab (HCRL)**, Aerospace Engineering & Engineering Mechanics, advised by Prof. Luis Sentis (Jan. 2018 – May 2021).
- Contributor to the UT Austin Villa@Home RoboCup team (AI-HRI 2019).

Research Experience

HIRO Group, University of Colorado Boulder — *Boulder, CO, USA*

Graduate Research Assistant — *NASA NSTGRO Fellow* — *Aug. 2021 – Present*

- Advised by Prof. Alessandro Roncone; research on fail-active manipulation, embodiment-aware policies, and contact-rich control.
- Pioneered a diffusion policy that recovers from any joint failure on a 7-DoF Franka Panda without retraining — **99.5%** task success vs. **42.4%** for RRT, **10/10** real-world tasks with multiple joints locked, trained on **4.7M** trajectories (ICRA 2026).
- Architected a fail-active controller for a two-arm package induction cell that reroutes work around faulty arms and conveyors instead of halting — a diffusion policy conditioned on a graph of each component's remaining capacity, giving **10**$\times$ the throughput of the graph-free policy under severe faults (ICRA 2027, under review).
- Devised non-prehensile pushing and poking primitives that complete tasks after an end-effector failure — reachable workspace up to **79%** larger, **88.9%** task success (IROS 2024).
- Co-developed a compliant controller that bounds the arm's energy during contact (IFAC World Congress 2026, oral) with the ROCC Lab (Prof. Marco M. Nicotra).
- Collaborate with NASA Johnson Space Center Robotics on long-term reliability and graceful degradation of robotic systems, supported by the NASA NSTGRO fellowship (Grant 80NSSC22K1211).

Human Centered Robotics Lab (HCRL), The University of Texas at Austin *Austin, TX, USA*
Undergraduate Researcher *Jan. 2018 – May 2021*

- Trained object-recognition models for the UT Austin Villa@Home service robot and implemented data-augmentation techniques that improved recognition from **60%** to $>$**95%** in the real world (RoboCup 2019).
- Built the robot's navigation, task planning, and behavior-tree task execution with Prof. Luis Sentis and Prof. Justin Hart; co-authored the team's AAAI AI-HRI 2019 report.

Professional Experience

Plus One Robotics *Boulder, CO, USA*
Robotics AI/ML Researcher *(Part-time)* *Jun. 2025 – Present*

- Delivered the proof of concept for learned motion planning on the package induction cell, targeting a cut in new-site spin-up from 8–12 weeks to 6–10 by removing most post-install tuning; the CTO reassigned $\sim$**6 engineers** to productionize it.
- Owned the fine-tuning of a vision-language-action (VLA) policy on 20 hours of test-cell data; after three months it runs the cell at **65%** of the classical system's throughput and retries failed picks without the hand-coded confirmation step the classical system needs.
- Established the policy architecture the team builds on — a scalable multi-model task-and-motion-planning approach grounded in a graph representation of the cell, so one policy runs across cells with different arms and conveyors.
- Built out a half-size closed-loop test cell and a five-conveyor, two-arm Isaac Sim model that has generated **150+** hours of training data; validated on the full-scale cell in San Antonio every two months.

Johns Hopkins University Applied Physics Laboratory *Laurel, MD, USA*
Robotics Software Engineer *(Intern)* *Jun. 2021 – Jul. 2021*

- Built a ROS 1 and ROS 2 simulator with Gazebo models of real buildings so the team could keep developing between tests on the physical robots.

METECS, NASA Johnson Space Center *Houston, TX, USA*
Robotics Software Intern *May 2020 – Aug. 2020*

- Wrote affordance templates — reusable task models of grasp points and motions — so Robonaut 2, NASA's International Space Station humanoid, could open latches and doors.

Publications

Bold denotes the author of this CV; underline denotes equal contribution. Authors are listed in the order they appear on the publication.

Conference Papers

1. **Gilberto G. Briscoe-Martinez**, Yaashia Gautam, Rahul Shetty, Anuj Pasricha, Marco M. Nicotra, and Alessandro Roncone, "Moving On, Even When You're Broken: Fail-Active Trajectory Generation via Diffusion Policies Conditioned on Embodiment and Task," in *IEEE International Conference on Robotics and Automation*, 2026.
2. Yaashia Gautam, **Gilberto Briscoe-Martinez**, Adhitya Mohan, Nataliya Nechyporenko, Alessandro Roncone, and Marco M. Nicotra, "Compliant Explicit Reference Governor for Contact Friendly Robotic Manipulators," in *International Federation of Automatic Control (IFAC) World Congress*, 2026.
3. **Gilberto Briscoe-Martinez**, Anuj Pasricha, Ava Abderezaei, Santosh Chaganti, Sarath Chandra Vajrala, Sri Kanth Popuri, and Alessandro Roncone, "Exploring How Non-Prehensile Manipulation Expands Capability in Robots Experiencing Multi-Joint Failure," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2024. [doi]
4. Rishi Shah, Yuqian Jiang, Haresh Karnan, **Gilberto Briscoe-Martinez**, Dominick Mulder, Ryan Gupta, Rachel Schlossman, Marika Murphy, Justin Hart, Luis Sentis, and Peter Stone, "Solving Service Robot Tasks: UT Austin Villa@Home 2019 Team Report," in *AAAI Fall Symposium on Artificial Intelligence and Human-Robot Interaction*, 2019. [pdf]

Under Submission

1. **Gilberto G. Briscoe-Martinez** and Alessandro Roncone, "What Can My Neighbor Still Do? Fail-Active Control for Process-Coupled Robot Cells via Graph-Conditioned Diffusion," submitted to *IEEE International Conference on Robotics and Automation*, 2027.
2. Ava Abderezaei, Nataliya Nechyporenko, Joseph Miceli, **Gilberto Briscoe-Martinez**, and Alessandro Roncone, "Physically Accurate Rigid-Body Dynamics in Particle-Based Simulation," submitted to *IEEE International Conference on Robotics and Automation*, 2027. [arXiv] [code]

Patents

1. Pawankumar Jajoo, Elizabeth R. Goodfriend, and **Gilberto G. Briscoe-Martinez**, "Computer-Implemented System and Method for Administering an Examination," U.S. Patent **US 10,339,827** (Issued). [link]

Awards & Fellowships

Plus One Industry Fellowship *$80,000*
Plus One Robotics *2026 – 2027*

- **Sole awardee in 2026** — no fellowship was awarded in 2025.

Black in Robotics Travel Award *$2,000*
Black in Robotics *2026*

NASA Space Technology Graduate Research Opportunity (NSTGRO) Fellowship *$336,000*
NASA — Grant 80NSSC22K1211 *2022 – 2026*

- Nationally competitive (∼50–65 awards nationwide per cycle); funded four years of PhD research in partnership with the Dexterous Robotics Lab at NASA Johnson Space Center.

Colorado Graduate Grant *$20,000*
University of Colorado Boulder *2022 – 2027*

Graduate Education for Minorities (GEM) Industry Fellowship *$20,000*
The National GEM Consortium *2021 – 2022*

- National fellowship pairing selected underrepresented STEM students with an industry sponsor.

Rejected Grant Proposals

xTechOverwatch Autonomous Systems Competition *U.S. Army*
Co-author · U.S. Army xTech Program *2025*

SMORES: A System for MOdular and REconfigurable RobotS for Long-term Lunar Surface Missions *NASA*
Co-author · NASA Innovative Advanced Concepts (NIAC) *2024*

- Advanced to the second round of review — approximately 30 of ∼300 submitted proposals (∼10%) were invited.

Service

Peer Review

2024	**Reviewer**, IEEE Transactions on Robotics (T-RO)	*Journal*
2025	**Reviewer**, IEEE Robotics and Automation Letters (RA-L)	*Journal*
2025	**Reviewer**, Robotics and Computer-Integrated Manufacturing	*Journal*
2022	**Reviewer**, IEEE/RSJ International Conference on Intelligent Robots and Systems	*Conference*
2026	**Reviewer**, IEEE/RSJ International Conference on Intelligent Robots and Systems	*Conference*
2026	**Reviewer**, Conference on Robot Learning (CoRL)	*Conference*
2026	**Reviewer**, IEEE International Conference on Robotics and Automation	*Conference*

Talks

2022–2024	**Invited Talk**, NASA Johnson Space Center	*Houston, TX*
2023, 2024	**Invited Talk**, Rocky Mountain AI Interest Group	*Boulder, CO*
2023	**Invited Talk**, Karman+	*Denver, CO*
2022	**Speaker**, Chancellor's Annual Dinner	*CU Boulder*

Outreach

2024, 2026	**Research Mentor**, Summer Multicultural Access to Research Training (SMART) Program	*CU Boulder*
2026	**Capstone Mentor**, Senior Capstone Design Course	*CU Boulder*
2026	**Panelist**, GEM GRAD Lab	*GEM Consortium*
Ongoing	**Volunteer**, Office of Graduate Access & Retention — recruiting and outreach events to broaden participation in research	*CU Boulder*

Mentorship

Students mentored:

- Rahul Shetty — *stealth startup*
- Santosh Chaganti
- Sarath Chandra Vajrala — *Figure*
- Sri Kanth Popuri — *Retina Robotics*
- Adhitya Mohan — *AMD*
- Liam Olzcak — *Amazon*
- Maneesh Gundluru
- Umair Yousaf
- Gyanig Kumar — *PhD, University of Wisconsin–Madison*
- Stephen Gillet
- Abraham Saleh
- Klara Nitsche
- Guilherme Valgas Schmidt
- Weston Brach
- Raul Castillo
- Macy Crow